

Curriculum Vitæ

Alexandre BOUSSE

1 Personal Details

First Name:	Alexandre
Family Name:	Bousse
Date of Birth:	13th of June, 1980
Place of Birth:	Rennes, France
Citizenship:	French
Position:	Full Researcher (<i>chargé de recherche</i>)
Laboratory:	Laboratoire de Traitement de l'Information Médicale (LaTIM) INSERM, UMR 1101, Brest, France
Email Address:	alexandre.bousse@univ-brest.fr
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2 Academic Career

2.1 Professional Experience

2021–present	Full Researcher (<i>chargé de recherche</i>), LaTIM, INSERM, UMR 1101, <i>Université de Bretagne Occidentale</i> (UBO), Brest, France
2018–2021	Associate Professor , LaTIM, INSERM, UMR 1101, UBO, Brest, France
2009–2018	Post-doctorate , Institute of Nuclear Medicine (INM), University College London (UCL), London, UK
2005–2008	PhD Candidate , <i>Laboratoire du Traitement du Signal et de l'Image</i> (LTSI), INSERM, UMR 1099, <i>Université de Rennes 1</i> , Rennes, France

2.2 Education

2019	Habilitation à diriger des recherches (<i>habilitation thesis</i>), LaTIM, INSERM, UMR 1101, UBO, Brest, France Title: “Contributions à la reconstruction tomographique compensée en mouvement” Viva: 07/10/2019
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	Jury: • Dimitris Visvikis, LaTIM, INSERM, UMR 1101, UBO, Brest, France • Françoise Pène, UBO, Brest, France • Claude Comtat, CEA, Orsay, France • Andrew Reader, King's College London, London, UK • Michel Defrise, <i>Université Libre de Bruxelles</i>, Brussels, Belgium
2005–2008	PhD, Signal Processing, LTSI, UMR 1099, <i>Université de Rennes 1</i>, Rennes, France, and Laboratory of Image Science and Technology (LIST), Southeast University (SEU), Nanjing, China Title: “Inverse Problems and Application to Motion-Compensated Rotational X-ray Angiography” Viva: 08/12/2008 Jury: • Directors: – J.-L. Coatrieux, LTSI, INSERM, UMR 1099, <i>Université de Rennes 1</i>, Rennes, France – H. Shu, LIST, SEU, Nanjing, China – C. Toumoulin, LTSI, INSERM, UMR 1099, <i>Université de Rennes 1</i>, Rennes, France • President: J. Demongeot, <i>Université Joseph Fourier</i>, Grenoble, France • Referees: – J. Yang, Nanjing University of Science and Technology, Nanjing, China – C. Roux, <i>Télécom Bretagne</i>, Brest, France • Reviewers: – L. Luo, LIST, SEU, Nanjing, China – D. Xia, Nanjing University of Science and Technology, Nanjing, China
2004–2005	Master of Science (DEA), Statistics , <i>Université de Rennes 1</i> , Rennes, France
2003–2004	Master of Science (DESS), Statistics , <i>Université de Rennes 1</i> , Rennes, France
1998–2003	Bachelor of Science, Mathematics , <i>Université de Rennes 1</i> , Rennes, France

3 Teaching Activities and Research Supervision

3.1 Teaching

2025–present	École d’Ingénieur Statistique Data Science et Big Data (ENSAI) (last year), Bruz, France Machine and deep learning 9-hour lecture per year.
2018–present	Institut Mines-Télécom (IMT) Atlantique (2nd year), Brest, France Advanced Medical Image Reconstruction 5-hour lecture per year & 6-hour lab per year Mathematics and algorithms for image reconstruction Machine and deep learning.
2018–2021	M2 Master Signaux Images en Biologie et Médecine (SIBM), UBO, Brest, France Introduction to Image Reconstruction 6-hour lecture per year (ended in 2021)

2018–2022	France: 12-hour lecture & 6-hour lab per year M2 physique et instrumentation , UBO, Brest Introduction to Image Reconstruction 12-hour lecture & 6-hour lab per year. (ended in 2022)
2018–2021	Coordinator of M2 SIBM at UBO
2018–2021	M1 biologie-santé , UBO, Brest, France Image Segmentation 6-hours lecture per year.
2018–2021	Medical and Dental School , UBO, Brest, France Pix 100 hours lab per year.
2004–2007	ENSAI (first year), Bruz, France Probability & Statistics (contractual teaching during PhD) Teaching topics: measure theory, random variables, parametric statistics, statistical hypothesis testing, linear regression 64-hour science lab in total.

3.2 PhD Supervision and Co-supervision

2024–present	Clémentine Phung-Ngoc —UBO Title: “Motion-corrected Low-dose PET/CT with Unsupervised Learning” Director: Alexandre Bousse (50%) Co-director: Olivier Saut (25%), Hong-Phuong Dang (25%) Publications: [PP1], [CP3] Status: ongoing
2024–present	Antoine De Paepe —UBO Title: “Motion-compensated CT Reconstruction with Unsupervised Learning” Director: Alexandre Bousse (100%) Publications: [J1], [CP2] Status: ongoing
2023–present	Thore Dassow —UBO Title: “Scored-Based Diffusion Models for Material Decomposition in PCCT” Director: Alexandre Bousse (100%) Publications: [CP1] (best Fully3D 2025 paper award) Status: ongoing
2022–present	Corentin Vazia — <i>Université Bretagne Sud</i> (UBS) Title: “Scored-Based Diffusion Models in Spectral CT” Director: Jacques Froment (50%) Co-director: Alexandre Bousse (50%) Publications: [PP2], [CP9], [CP10] Status: ongoing
2021–2025	Juan José Molina —Pontificia Universidad Católica de Chile (PUC) Title: “Deep Learning Techniques for T_2^* Image Reconstruction” Director: Matias Courdurier (80%) Co-supervisor: Alexandre Bousse (20%) Publications: [PP5] Status: completed

2021–2025	Youness Mellak—UBO Title: “Deep Learning Techniques for 3-gamma PET Imaging” Director: Dimitris Visvikis (25%) Co-director: Alexandre Bousse (75%) Publications: [J3], [J4], [CP6]–[CP8], [CP16] Status: completed
2021–2025	Valentin Gautier—Institut National des Sciences Appliquées (INSA) Lyon Title: “Reconstruction bimodale d’images TEP/IRM assistée par intelligence artificielle” Director: Bruno Sixou (50%) Co-supervisor: Alexandre Bousse (25%), Voichita Maxim (25%) Publications: [J6], [CP5], [CP12] Status: completed
2021–2024	Noel Pinton—UBO Title: “Synergistic PET/CT Reconstruction using Deep Learning” Director: Alexandre Bousse (100%) Publications: [J5], [CP13], [CP14] Status: completed
2021–2024	Zhihan Wang—UBO Title: “Spectral computed tomographic image reconstruction using deep learning” Director: Alexandre Bousse (75%) Co-director: Frank Vermet (25%) Publications: [CP4], [J7], [CP17] Status: completed
2017–2022	Baptiste Laurent—UBO Title: “Estimation des diffusés en TEP par apprentissage profond” Director: Nicolas Boussion Co-director: Alexandre Bousse (100%) Publications: [J2], [J8], [CP23] Status: completed
2018–2022	Sai Sundar Kandarpa—UBO Title: “Tomographic Image Reconstruction with Direct Neural Network Approaches” Director: Alexandre Bousse (100%) Publications: [RV1], [RV2], [CP15], [J9], [J13], [CP20], [CP29] Status: completed
2018–2022	Suxer Alfonso Garcia—UBO Title: “Multi-channel Computed Tomographic Image Reconstruction by Exploiting Structural Similarities” Director: Alexandre Bousse (100%) Publications: [J10], [CP18], [CP19], [CP21] Status: completed
2017–2020	Debora Giovagnoli—IMT Title: “3-γ Image Reconstruction using LXe Compton Camera XEMIS2” Director: Dimitris Visvikis Co-director: Alexandre Bousse (50%) Publications: [J12] Status: completed

2016–2020	Ludovica Brusafferri —UCL Title: “Improving Quantification in non-TOF 3D PET/MR by Incorporating Photon Energy Information” Director: Kris Thielemans (50%) Co-supervisor: Alexnadre Bousse (50%) Publications: [J11], [J16], [CP22], [CP32], [CP35] Status: completed
2016–2020	Élise Émond —UCL Title: “Improving Quantification in Lung PET/CT for the Evaluation of Disease Progression and Treatment Effectiveness” Director: Kris Thielemans (50%) Co-supervisor: Alexnadre Bousse (50%) Publications: [J17], [J18], [CP25]–[CP27] Status: completed
2014–2018	Yu-Jung Tsai —UCL Title: “Penalised Image Reconstruction Algorithms for Efficient and Consistent Quantification in Emission Tomography” Director: Kris Thielemans (50%) Co-supervisor: Alexnadre Bousse (50%) Publications: [J14], [J19], [CP33], [J21], [CP36], [CP39], [CP42] Status: completed
2010–2015	Sarah Cade —UCL Title: “Attenuation Correction of Myocardial Perfusion Scintigraphy Images without Transmission Scanning” Director: Brian F. Hutton (50%) Co-supervisor: Alexnadre Bousse (50%) Publications: [CP58] Status: completed

3.3 Master Students Supervision

2024	Antoine De Paepe, Clementine Phung-Ngoc, Apolline Guerineau (ENSAI) “Compressed-sensing for CT reconstruction”
2023	Tiban Dorel (IMT Atlantique) “CT reconstruction with unrolling architectures”
2022	Mariana Yuli Sato do Nascimento (IMT Atlantique) “PET/CT denoising with conditional GANs”
2019	Celia Boutalbi (Université de Bordeaux) “Direct material decomposition in dual-energy CT”

4 Grants and External Funding

4.1 Funding at Current Position (since 01/09/2018)

2025–present	Projets Intra- et Inter-Instituts Brestois (IB) de l’UBO (France) Amount awarded: €6,000 Role: PI
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2024–present	Project Title: “Diffusion Spectrale” Inserm-Inria PhD grant (France) Amount awarded: €130,000 Role: PI
2024–present	Project Title: “Motion-corrected Low-dose PET/CT with Unsupervised Learning” Région Bretagne / UBO PhD grant (France) Amount awarded: €130,000 Role: PI
2023–present	Project Title: “Motion-compensated CT Reconstruction with Unsupervised Learning” France Life Imaging WP4 (France) Amount awarded: €20,000 Role: PI
2020–2024	Project Title: “Deepmulti: Deep Learning Techniques for Multimodal Image Reconstruction” ANR – ANR-20-CE45-0020 (France) Amount awarded: €496,800 Role: PI
2019–2022	Project Title: “MultiRecon: Machine-Learning for Multimodal Medical Image Reconstruction” France Life Imaging WP4 (France) Amount awarded: €24,000 Role: PI
2019–2022	Project Title: “Dual-Tracer in Dynamic PET” (complement funding to the <i>Émergence</i> project) AO Émergence Cancéropôle Grand Ouest (France) Amount awarded: €15,000 Role: PI Project Title: “Dual-Tracer in Dynamic PET”

4.2 Funding during Postdoc at UCL (2009–2018)

2016–2018	GE Healthcare (USA) Amount awarded: \$150,000 Role: co-PI PI: Kris Thielemans Project Title: “Motion-Compensated PET/CT”
2013–2015	FP7 – HEALTH Program 305311 (EU) Amount awarded: €575,622 (€5,981,463 in total for all the EU partners) Role: Research Fellow PI: Brian F. Hutton Project Title: “Development of an integrated SPECT/MRI system”
2013–2016	EPSRC – EP/K005278/1 (UK) Amount awarded: £1,274,298 Role: WP leader PI: Brian F. Hutton Project Title: “Exploiting the Unique Quantitative Capabilities Offered by Simultaneous PET/MRI”
2009–2012	EPSRC – EP/G026483/1 (UK)

Amount awarded: £767,088	Amount awarded: £767,088
Role: Research Fellow	Role: Research Fellow
PI: Brian F. Hutton	PI: Brian F. Hutton
Project Title: "Optimising Reconstruction to Accommodate Complex System Models for Emission Tomography"	Project Title: "Optimising Reconstruction to Accommodate Complex System Models for Emission Tomography"

5 Academic Service and Scientific Diffusion

5.1 Meetings Organisation

5.1.1 Conferences

2022–present	Member of IEEE Nuclear and Medical Imaging Sciences Council
2023	Session Chairman , IEEE Medical Imaging Conference, Vancouver, Canada
2022	Session Chairman , IEEE Medical Imaging Conference, Milan, Italy
2021	Student Paper Awards Committee , International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine
2018	Session Chairman , IEEE Medical Imaging Conference, Sydney, Australia

5.1.2 Workshops

2022	AI Wild West (organiser) https://wildwestworkshop.github.io/
2022	Rencontre Lyonnaise en Imagerie d'Émission (organiser) https://emilyworkshop.github.io/

5.2 Scientific Evaluation

5.2.1 Peer Reviewing

Journal Associate Editor

- IEEE TRPMS
- Frontiers in Medicine

Journal Peer Review

- IEEE Transactions on Medical Imaging
- IEEE Transactions on Biomedical Engineering
- IEEE Transactions on Radiation and Plasma Medical Sciences
- IEEE Transactions on Computational Imaging
- Physics in Medicine and Biology
- Neuroimage
- PLOS one
- Philosophical Transactions A
- Frontiers in Nuclear Medicine
- European Journal of Nuclear Medicine and Molecular Imaging

Grant Peer Review

- Nantes Excellence Trajectory (NEXT)
- *Wetenschappelijk Fonds Willy Gepts* (WFWG)
- Netherlands Organisation for Scientific Research (NWO)
- Expertise Scientifique Cifre

Conference Committee

- MICCAI 2020, 2021, 2022, 2023, 2024, 2025
- Fully 3D 2021, 2023, 2025
- IEEE Nuclear Science Symposium and Medical Imaging Conference 2014, 2015, 2016, 2017, 2018, 2019, 2021, 2022, 2023, 2024

5.2.2 PhD Examination

2023	Kannara Mom , “Deep learning-based phase retrieval for X-ray phase contrast imaging”, INSA Lyon, Lyon, France.
2023	Guillaume Corda , “Joint PET-MR deep learned reconstruction”, King’s College London, London, UK.
2022	Louise Friot , “ <i>Méthodes de reconstruction avancées en tomographie dentaire par faisceau conique</i> ”, INSA Lyon, Lyon, France.
2021	Zacharias Chalampalakis , “Modelling and Reconstruction of Whole-Body Parametric Maps in PET-MRI Pharmacological Imaging”, Biomaps, Orsay, France.

5.2.3 PhD Jury

2024	Noel Pinton , “Synergistic PET/CT Reconstruction using Deep Learning”, UBO, Brest, France.
2024	Zhihan Wang , “Spectral computed tomographic image reconstruction using deep learning”, UBO, Brest, France.
2022	Suxer Alfonso Garcia , “Multi-channel computed tomographic image reconstruction by exploiting structural similarities”, UBO, Brest, France.
2022	Sai Sundar Kandarpa , “Tomographic image reconstruction with direct neural network approaches”, UBO, Brest, France.
2020	Debora Giovagnoli , “3- γ Image Reconstruction using LXe Compton Camera XEMIS2”, UBO, Brest, France.

5.2.4 Other Jurys

2022–present	Jury contrat doctoral établissement (CDE), école doctorale biologie santé, UBO.
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5.3 Recent Invited Talks

Jul. 2024	“Tomodensitométrie à comptage photonique : projets du LaTIM en reconstruction d’image”, LaTIM Day , LaTIM, Brest, France.
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Nov. 2023	"Modèles génératifs pour reconstruction multimodale", GDR ISIS, INSA, Lyon, France.
Mar. 2022	"Generative Models for Multichannel Image Reconstruction", <i>Pontificia Universidad Católica de Chile</i> , Santiago, Chile.
Oct. 2021	"Multirecon: Machine Learning for Multimodal Medical Image Reconstruction", INSA, Lyon, France.
Jul. 2019	"Innovations in Image Reconstruction", LaTIM, Brest, France.
Feb. 2019	"Reconstruction d'image en tomographie à émission de positons par maximum de vraisemblance avec compensation du mouvement respiratoire", <i>Laboratoire de Mathématiques de Bretagne Atlantique</i> , Brest, France.
Jun. 2018	"Respiratory Motion Correction in PET/CT and PET/MR", Mathematical Methods for Spatiotemporal Imaging, SIAM Conference on Imaging Science 2018, Bologna, Italy.
Mar. 2017	"Maximum-Likelihood PET Reconstruction and Motion Estimation", <i>Pontificia Universidad Católica de Chile</i> , Santiago, Chile.
Sept. 2016	"Direct Motion Compensation in Attenuation-Corrected PET/CT and PET/MR Reconstruction", UCL PET/MR Methods Symposium, London, UK.
May 2016	"Reconstruction en PET-CT avec compensation du mouvement par techniques de maximum de vraisemblance", CEA, Orsay, France.
Mar. 2016	"Motion-Compensated PET Image Reconstruction by Maximum-Likelihood", Newton Project Workshop on Brazil/UK Collaboration: the Future of Molecular Imaging, Recife, Brazil.
Nov. 2015	"Gated PET Reconstruction with Motion Compensation and Attenuation Correction using non-Gated CT", Brain Institute, <i>Hospital Israelita Albert Einstein</i> , São Paulo, Brazil.

5.4 Software Development: JRM

Name	Joint Reconstruction and Motion estimation (JRM)
Language	Matlab/C++
Description	Joint Reconstruction and Motion estimation (JRM) is a toolbox for motion-compensated attenuation-corrected PET reconstruction that I developed for UCL and GE Healthcare. While the full version cannot be distributed, a "light" version is available at the address below.
Source code	https://gitlab.com/abousse/jrm_lite

6 Publications

Journal Articles

- [J1] A. De Paepe, **A. Bousse**, C. Phung-Ngoc, Y. Mellak, and D. Visvikis, "Adaptive diffusion models for sparse-view motion-corrected head cone-beam CT," *IEEE Transactions on Radiation and*

- Plasma Medical Sciences*, 2025. doi: [10.1109/TRPMS.2025.3637124](https://doi.org/10.1109/TRPMS.2025.3637124). [Online]. Available: <https://arxiv.org/abs/2504.14033>.
- [J2] B. Laurent, **A. Bousse**, T. Merlin, A. Rominger, K. Shi, and D. Visvikis, "Evaluation of deep learning-based scatter correction on a long-axial field-of-view PET scanner," *European Journal of Nuclear Medicine and Molecular Imaging*, vol. 52, no. 7, pp. 2563–2576, 2025. doi: [10.1007/s00259-025-07120-6](https://doi.org/10.1007/s00259-025-07120-6). [Online]. Available: <https://arxiv.org/pdf/2501.01341>.
 - [J3] Y. Mellak, **A. Bousse**, T. Merlin, É. Émond, and D. Visvikis, "Dual-input dynamic convolution for positron range correction in PET image reconstruction," *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2025. doi: [10.1109/TRPMS.2025.3647264](https://doi.org/10.1109/TRPMS.2025.3647264). [Online]. Available: <https://arxiv.org/abs/2503.00587>.
 - [J4] Y. Mellak, **A. Bousse**, T. Merlin, D. Giovagnoli, and D. Visvikis, "Direct3 γ pet: A pipeline for direct three-gamma PET image reconstruction," *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2025. doi: [10.1109/TRPMS.2025.3577810](https://doi.org/10.1109/TRPMS.2025.3577810). [Online]. Available: <https://arxiv.org/pdf/2407.18337>.
 - [J5] N. J. Pinton, **A. Bousse**, C. Cheze-Le Rest, and D. Visvikis, "Multibranch generative models for multichannel imaging with an application to PET/CT synergistic reconstruction," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 9, no. 5, pp. 654–666, 2025. doi: [10.1109/TRPMS.2025.3532176](https://doi.org/10.1109/TRPMS.2025.3532176). [Online]. Available: <https://arxiv.org/abs/2404.08748>.
 - [J6] V. Gautier, **A. Bousse**, F. Sureau, C. Comtat, V. Maxim, and B. Sixou, "Bimodal PET/MRI generative reconstruction based on VAE architectures," *Physics in Medicine & Biology*, vol. 69, 2024. doi: [10.1088/1361-6560/ad9133](https://doi.org/10.1088/1361-6560/ad9133). [Online]. Available: <https://iopscience.iop.org/article/10.1088/1361-6560/ad9133>.
 - [J7] Z. Wang, **A. Bousse**, F. Vermet, J. Froment, B. Vedel, A. Perelli, J.-P. Tasu, and D. Visvikis, "Uconnect: Synergistic spectral CT reconstruction with U-Nets connecting the energy bins," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 2, pp. 222–233, 2024. doi: [10.1109/TRPMS.2023.3330045](https://doi.org/10.1109/TRPMS.2023.3330045). [Online]. Available: <https://arxiv.org/abs/2311.00666>.
 - [J8] B. Laurent, **A. Bousse**, T. Merlin, S. Nekolla, and D. Visvikis, "PET scatter estimation using deep learning U-net architecture," *Physics in Medicine & Biology*, vol. 68, no. 6, p. 065 004, 2023. doi: [10.1088/1361-6560/ac9a97](https://doi.org/10.1088/1361-6560/ac9a97).
 - [J9] V. S. S. Kandarpa, A. Perelli, **A. Bousse**, and D. Visvikis, "LRR-CED: Low-resolution reconstruction-aware convolutional encoder-decoder network for direct sparse-view CT image reconstruction," *Physics in Medicine & Biology*, vol. 67, no. 15, p. 155 007, 2022. doi: [10.1088/1361-6560/ac7bce](https://doi.org/10.1088/1361-6560/ac7bce).
 - [J10] A. Perelli, S. L. Alfonso Garcia, **A. Bousse**, J.-P. Tasu, N. Efthimiadis, and D. Visvikis, "Multi-channel convolutional analysis operator learning for dual-energy CT reconstruction," *Physics in Medicine & Biology*, vol. 67, no. 6, p. 065 001, 2022. doi: [10.1088/1361-6560/ac4c32](https://doi.org/10.1088/1361-6560/ac4c32). [Online]. Available: <https://arxiv.org/abs/2203.05968v1>.
 - [J11] L. Brusaferri, É. C. Émond, **A. Bousse**, R. Twyman, A. C. Whitehead, D. Atkinson, S. Ourselin, B. F. Hutton, S. Arridge, and K. Thielemans, "Detection efficiency modelling and joint activity and attenuation reconstruction in non-TOF 3D PET from multiple-energy window data," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 6, no. 1, pp. 87–97, 2021. doi: [10.1109/TRPMS.2021.3064239](https://doi.org/10.1109/TRPMS.2021.3064239).
 - [J12] D. Giovagnoli, **A. Bousse**, N. Beaupere, C. Canot, J.-P. Cussonneau, S. Diglio, A. Iborra Carreres, J. Masbou, T. Merlin, E. Morteau, Y. Xing, Y. Zhu, D. Thers, and D. Visvikis, "A pseudo-TOF image reconstruction approach for three-gamma small animal imaging," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 6, pp. 826–834, 2021. doi: [10.1109/TRPMS.2020.3046409](https://doi.org/10.1109/TRPMS.2020.3046409).

- [J13] V. S. S. Kandarpa, **A. Bousse**, D. Benoit, and D. Visvikis, "DUG-RECON: A framework for direct image reconstruction using convolutional generative networks," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 1, pp. 44–53, 2021. doi: [10.1109/TRPMS.2020.3033172](https://doi.org/10.1109/TRPMS.2020.3033172). [Online]. Available: <https://arxiv.org/abs/2012.02000>.
- [J14] Y.-J. Tsai, **A. Bousse**, S. Arridge, C. W. Stearns, B. F. Hutton, and K. Thielemans, "Penalized PET/CT reconstruction algorithms with automatic realignment for anatomical priors," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 3, pp. 362–372, 2021. doi: [10.1109/TRPMS.2020.3025540](https://doi.org/10.1109/TRPMS.2020.3025540). [Online]. Available: <https://arxiv.org/abs/1911.08012>.
- [J15] **A. Bousse**, M. Courdurier, É. C. Émond, K. Thielemans, B. F. Hutton, P. Irarrazaval, and D. Visvikis, "PET reconstruction with non-negativity constraints in projection space: Optimization through hypo-convergence," *IEEE Transactions on Medical Imaging*, vol. 39, no. 1, pp. 75–86, 2020. doi: [10.1109/TMI.2019.2920109](https://doi.org/10.1109/TMI.2019.2920109). [Online]. Available: <https://hal.archives-ouvertes.fr/hal-02144923>.
- [J16] L. Brusaferri, **A. Bousse**, É. C. Émond, R. Brown, Y.-J. Tsai, D. Atkinson, S. Ourselin, C. C. Watson, B. F. Hutton, S. Arridge, et al., "Joint activity, attenuation and scatter estimation from multiple energy window data in non-TOF 3D PET: A preliminary study," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 4, no. 4, pp. 410–421, 2020. doi: [10.1109/TRPMS.2020.2978449](https://doi.org/10.1109/TRPMS.2020.2978449). [Online]. Available: <https://ieeexplore.ieee.org/document/9024002>.
- [J17] É. C. Émond, **A. Bousse**, L. Brusaferri, B. F. Hutton, and K. Thielemans, "Improved PET/CT respiratory motion compensation by incorporating changes in lung density," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 4, no. 5, pp. 594–602, 2020. doi: [10.1109/TRPMS.2020.3001094](https://doi.org/10.1109/TRPMS.2020.3001094). [Online]. Available: <https://ieeexplore.ieee.org/document/9112356>.
- [J18] É. C. Émond, **A. Bousse**, M. Machado, J. Porter, A. M. Groves, B. F. Hutton, and K. Thielemans, "Effect of attenuation mismatches in time of flight PET reconstruction," *Physics in Medicine & Biology*, vol. 65, no. 8, p. 085 009, 2020. doi: [10.1088/1361-6560/ab7a6f](https://doi.org/10.1088/1361-6560/ab7a6f). [Online]. Available: <https://iopscience.iop.org/article/10.1088/1361-6560/ab7a6f>.
- [J19] Y.-J. Tsai, G. Schramm, S. Ahn, **A. Bousse**, S. Arridge, J. Nuyts, B. F. Hutton, C. W. Stearns, and K. Thielemans, "Benefits of using a spatially-variant penalty strength with anatomical priors in PET reconstruction," *IEEE Transactions on Medical Imaging*, vol. 39, no. 1, pp. 11–22, 2020. doi: [10.1109/TMI.2019.2913889](https://doi.org/10.1109/TMI.2019.2913889).
- [J20] A. Iborra, A. J. González, A. Gonzalez-Montoro, **A. Bousse**, and D. Visvikis, "Ensemble of neural networks for 3D position estimation in monolithic PET detectors," *Physics in Medicine & Biology*, vol. 64, no. 19, p. 195 010, 2019. doi: [10.1088/1361-6560/ab3b86](https://doi.org/10.1088/1361-6560/ab3b86).
- [J21] Y.-J. Tsai, **A. Bousse**, M. J. Ehrhardt, C. W. Stearns, S. Ahn, B. F. Hutton, S. Arridge, and K. Thielemans, "Fast quasi-newton algorithms for penalized reconstruction in emission tomography and further improvements via preconditioning," *IEEE Transactions on Medical Imaging*, vol. 37, no. 4, pp. 1000–1010, 2018. doi: [10.1109/TMI.2017.2786865](https://doi.org/10.1109/TMI.2017.2786865). [Online]. Available: <https://doi.org/10.1109/TMI.2017.2786865>.
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